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SUPERSEDING
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MILITARY SPECIFICATION

CRUSHER, JAW, ROCK, AIRBORNE, 15-TPH CAPACITY

1. SCOPE

1.1 This specification covers a gasoline-engine-driven, single-pass, jaw-type, 15-ton-per-hour, airborne rock crusher, mounted on a 2-axle, single-tire, wagon-type trailer.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

FF-B-171	- Bearings, Ball, Annular (General Purpose).
FF-B-185	- Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
FF-B-187	- Bearing, Roller, Tapered.
QQ-S-681	- Steel Castings.
QQ-S-781	- Steel, Strapping, Flat.
QQ-S-790	- Steel Strapping, Round (Bare and Zinc-Coated).
SS-R-406	- Road and Paving Materials; Methods of Sampling and Testing.
TT-P-664	- Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
UU-T-81	- Tags, Shipping and Stock.
ZZ-B-206	- Belting, Flat; Conveyor, Rubber.
ZZ-I-550	- Inner Tube, Pneumatic Tire.
GGG-J-51	- Jacks, Mechanical and Hydraulic, Hand Operated.
KKK-L-370	- Lining, Friction (Clutch and Brake; Metallic, Metal-Ceramic and Semimetallic).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

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Military

MIL-P-116 - Preservation, Methods of.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (for the Treatment of Communications, Electronic, and Associated Electrical Equipment).
MIL-P-514 - Plates; Identification; Transportation-Data, and Blank.
MIL-I-631 - Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-L-2105 - Lubricating Oil, Gear, Multipurpose.
MIL-F-3541 - Fittings, Lubrication.
MIL-G-3859 - Grease Guns, Hand, High-Pressure, Lever-Operated (15-Ounce and 21-Ounce Capacity).
MIL-H-3992 - Hose and Hose Assembly, Rubber; Air and Vacuum Brake, Automotive.
MIL-L-4387 - Lubrication Dispensing Equipment Accessories
MIL-S-8660 - Silicone Compound.
MIL-E-10062 - Engines: Preparation for Shipment and Storage of.
MIL-G-10924 - Grease, Automotive and Artillery.
MIL-E-11275 - Engines, Gasoline, Industrial Type, General Specification for.
MIL-R-12144 - Reflector, Indicating, Clearance.
MIL-C-13486 - Cable, Special Purpose, Electrical: Low-Tension, Heavy-Duty Single-Conductor and Multi-Conductor.
MIL-I-15126 - Insulation Tape, Electrical, Pressure Sensitive Adhesive and Pressure Sensitive Thermosetting Adhesive.
MIL-H-15424 - Hand Tools, Packaging of.
MIL-C-16173 - Corrosion Preventive Compound, Solvent Cutback, Cold-Application.
MIL-L-21260 - Lubricating Oil, Internal-Combustion Engine, Preservative.
MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-C-52058 - Chains, Roller, Power Transmission, and Conveyor.
MIL-A-52363 - Air Cleaners, Intake: Dry-Type (for Internal Combustion Engine).
MIL-E-55301 - Electromagnetic Compatibility.

STANDARDS

Federal

FED. STD. No. 595 - Colors.

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Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U. S. Military Property.
MIL-STD-281	- Automobiles, Trucks, Truck - Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-421	- Chain, Roller; Power Transmission and Conveyor, Flat Link Plates, Single Pitch, Single and Multiple Strand: Connecting Links and Attachment Links.
MIL-STD-423	- Chain, Roller; Conveyor, Standard and Large Rollers, Flat Link Plates, Double Pitch, Single Strand; Connecting Links and Attachment Links.
MIL-STD-424	- Chain, Roller; Power Transmission, Offset Side Bars, Single-Pitch.
MIL-STD-642	- Identification Marking of Administrative, Combat, and Tactical Transport Vehicles.
MIL-STD-1400	- Engines, Gasoline or Diesel, Methods of Test.
MS16961	- Troughing Carrier Assembly, 4 In. Dia Roll, 4-5/8 In. Backing; 24 and 30 In. Belt Width.
MS16963	- Troughing Carrier Assembly, Impact, 4 In. Dia Rubber Roll, 4-5/8 In. Backing, 24 and 30 In. Belt Width.
MS16964	- Return Roll Assembly, 4 In. Dia Roll, for 4-5/8 In. Backing Trough Carrier; 24 and 30 In. Belt Width.
MS21322	- Clamps, Loop - Aluminum-Plain and Cushioned.
MS21333	- Clamps, Loop - Steel-Plain and Cushioned.
MS21334	- Clamps, Loop - Double, Cable or Tube, Steel, Plain and Cushioned.
MS25281	- Clamp, Loop - Plastic, Wire Support.
MS27142	- Connector, Plug, Electrical - Pin Contact, No. 12, 14, and 16 A.W.G., Waterproof.
MS27143	- Connector, Plug, Electrical - Socket Contact, No. 12 A.W.G., Waterproof.
MS27144	- Connector, Plug, Electrical - Socket Contact, No. 14 and 16 A.W.G., Waterproof.
MS27145	- Connector, Plug, Electrical - No. 14 A.W.G. Double Contact Waterproof.

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MS27146	- Connector, Receptacle, Electrical - No. 12, 14 and 16 A.W.G., Double Contact, Waterproof.
MS27147	- Adapter, Connector - 3 Way, No. 12, 14 and 16 A.W.G., Waterproof.
MS27148	- Pin Contact, Electrical Connector - No. 12, 14 and 16 A.W.G.
MS33800	- Connector, Receptacle, Electrical - Pin Contact No. 12, 14 and 16 A.W.G., Waterproof.
MS35000	- Battery, Storage, Lead-Acid, Waterproof.
MS35388	- Tire, Pneumatic, Passenger Car, Light Truck, Truck and Bus.
MS35392	- Inner Tube, Pneumatic Tire.
MS35423	- Light, Marker, Clearance-Service.
MS35424	- Light, Marker, Clearance-Blackout.
MS35647	- Padlocks, Key Operated.
MS35746	- Coupling, Automotive - Air Brake Lines.
MS35773	- Radiators, Engine Cooling Industrial.
MS35829	- Hinge, Continuous (Piano), .075 Thickness, .187 and .250 Pin Dia.
MS35844	- Plug, Machine Thread; Magnetic (Drain).
MS35865	- Axle, Vehicular, Non-Driving, Single, for Tactical Wheeled Vehicles.
MS39020	- Band, Marker, Blank.
MS39062	- Circuit Breakers: Thermal, 30 Volt DC, Automatic, Waterproof, 15 to 30 Amp.
MS49006	- Plugs, Pipe, Magnetic - Headless; Aluminum, Iron or Steel.
MS51330	- Stop Light - Taillight, Vehicular - 24 Volt, Blackout Tail, Blackout Stop.
MS51337	- Coupler, Drawbar, Ring - Heavy-Duty.
MS53004	- Valve, Relay, Emergency - Breakaway, Air Brake.
MS53007	- Plate, Identification - Emergency and Service.
MS53044	- Wheel, Pneumatic Tire, Disk Type, with Ring, for Tactical Wheeled Vehicles.
MS53063	- Indicator, Air Cleaner, Intake; Restriction Mechanical Type.
MS75020	- Connector, Plug, Electrical - 12 Contact, Intervehicular, 24 Volt, Waterproof.
MS75021	- Connector, Receptacle, Electrical 12 Contact, Intervehicular, 24 Volt, Waterproof.
MS500010	- Truck, Dump, 5 Ton, 6 x 6, M51.

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DRAWINGS

Army - MO (ATAC)

8347201 - Schematic, Wiring Diagram, Semi-Type Trailer (Typical) 24 Volt.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents from a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code for Unfired Pressure Vessels, Section IX, Welding Qualifications

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N. Y. 10017.)

AMERICAN STANDARDS ASSOCIATION

B1.5	- Acme Screw Threads.
B4.1	- Preferred Limits and Fits for Cylindrical Parts.
B6.6	- Gear Tolerances and Inspection.
B55.1	- Specifications for Multiple V-Belt Drives.

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N. Y. 10017.)

RUBBER MANUFACTURERS ASSOCIATION, INC.

Engineering Standard for Drives Using Narrow V-Belts (3V, 5V, and 8V Cross Sections).

(Application for copies should be addressed to the Rubber Manufacturers Association, 444 Madison Avenue, New York 22, New York.)

AMERICAN WELDING SOCIETY

Standard Qualification Procedure.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N. Y. 10017.)

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AMERICAN TRUCKING ASSOCIATION, INC.

Recommended Equipment Specification E-3-1961.
National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, 1616 P St., NW, Washington, D. C. 20036.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N.Y. 10016.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 485 Lexington Avenue, New York, N.Y. 10017.)

INTERSTATE COMMERCE COMMISSION

Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A128-60	- Austenitic Manganese Steel Castings.
A193-61T	- Alloy-Steel, Bolting Materials, for High Temperature Service.
A242	- Standard Specification for High Strength Low Alloy Structural Steel.
A536	- Ductile Iron Castings.
C170	- Standard Method of Test for Compressive Strength of Natural Building Stone.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

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3. REQUIREMENTS

3.1 Description. The crusher shall consist of a chassis upon which is mounted a chute-type feeder, a jaw-type crushing unit, a discharge conveyor, a power unit, and all necessary auxiliary fastenings, mountings, drives, guards, and platforms. The crusher shall be of the full trailer type, mounted on four pneumatic-tired wheels. The crusher shall conform to the following weight and dimensional requirements without sectionalization (except as permitted in 3.1(e)):

(a) Gross vehicle weight (GVW) shall be not more than 20,000 pounds.
(b) Overall length shall be not more than 420 inches with towbar horizontal end in straight-ahead position.
(c) Reducible length shall be not more than 345 inches with the front axle turned 180 degrees from the straight-ahead towing position and the towbar under the crusher.
(d) Overall width shall be not more than 100 inches.
(e) Overall height shall be not more than 92 inches with the discharge end of the conveyor folded down as specified in 3.13 and platform handrails removed as specified in 3.14.1.

3.1.1 Capacity. When crushing quarry rock as specified in 3.1.1.1 requiring 80 percent crushing, the jaw crusher shall have a capacity of not less than 15 tons per hour of 1-1/2 inch minus material with not less than 85 percent passing 1-1/2 inch-square screen openings.

3.1.1.1 Quarry rock. The quarry rock shall be diabase or traprock having an average compressive strength of not less than 32,000 pounds per square inch (psi) when tested in accordance with ASTM C170 and shall have a percentage of wear of not more than 20 when tested in accordance with SS-R-406, method 208.11.

3.1.2 Stress limitations. The crusher shall be constructed so that the maximum stress imposed on any structural component shall not exceed 60 percent of the yield point.

3.2 Preproduction model. The supplier shall furnish a crusher within the time frame specified in the contract or purchase order to prove, prior to starting production, that his production methods and choice of design detail will produce crushers that comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the preproduction model shall be subject to the approval of the contracting officer. Approval of the preproduction model by the activity concerned shall not relieve the supplier of his obligation to furnish crushers conforming to this specification.

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3.3 Initial production. Unless otherwise specified (see 6.2), the supplier shall furnish to the Government one or more crusher(s) (see 6.2) for testing as specified in 4.4.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Material not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.5 Safety. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded. All moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions. The crusher shall conform to all large trailer requirements of the ICC Motor Carrier Safety Regulations, parts 190 and 193, except as specified herein.

3.6 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments not normally removed. Covers or plates such as those on shrouds or housings which must be removed for adjustment or maintenance shall be equipped with quick-disconnect fastenings. Disconnect plugs and receptacles, multiple-line connectors in the electrical system, and detachable- and attachable-type fittings in hydraulic and pneumatic systems, as applicable, shall be utilized where practicable to permit component removal and replacement. All disconnect points shall be clearly indicated. Maximum practicable use shall be made of interchangeable hardware and fastening devices with a minimum number of types and sizes of bolts, screws, nuts, washers, and similar common (standard) parts. Lockwashers shall be secured to bolts or screws except where this practice is not practicable. Drain outlets shall be located for accessibility and shall permit the use of receptacles for collecting drainage. Drain plugs shall be removable by common wrench. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.7 Transportability performance. The crusher shall conform to the transport performance characteristics specified herein without malfunction, permanent deformation, or damage when lubricated, serviced and prepared for transport and towed by the truck specified in 3.15.

3.7.1 Highway operation. The crusher shall travel over smooth, hard-surfaced, primary highways at sustained speeds up to 25 mph.

3.7.2 Cross-country operation. The crusher shall travel over unimproved roads, open fields, rolling hills, and rough cross-country terrain at speeds up to 15 mph. The crusher shall travel on side slopes up to 30 percent. The crusher suspension shall have sufficient articulation

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to permit the crusher to roll over 8 inch high obstacles with any two wheels without loss of tire-to-surface contact of any wheel. The ground clearance of the crusher shall be not less than 12 inches, excluding axles.

3.7.3 Trailing and turning ability. The crusher shall trail the towing truck without weaving or sidesway. The crusher shall follow the truck without exceeding 75 percent of the ICC towed vehicle tracking tolerance (see 3.5), when towed over paved surfaces. Turns up to an angle of 60 degrees from straight-ahead, to the right and to the left shall not cause damage to nor interference between the drawbar of the crusher and the Military trucks. Jack-knife turns up to an angle of 110 degrees from straight-ahead, to the right and to the left shall not cause damage to nor interference between the crusher and the truck when the dolly is straight with the truck.

3.7.4 Braking ability. The combined service brakes of the truck and the crusher shall control, decelerate, and stop the combination in accordance with 3.5. The emergency breakaway automatic brake actuation shall perform in accordance with 3.5. The brakes shall be balanced to provide greater braking effort on the rear brakes then on the front brakes, to minimize jack-knifing tendencies.

3.7.5 Fording. The crusher, without additional fording equipment, shall ford hard-bottom water crossings to a depth of not less than 30 inches up to 15 minutes duration without damage. Immediately after fording, the brakes shall not exhibit water-fade, grabbing, or chatter.

3.8 Hoppers and chutes. All hoppers and chutes shall prevent spillage, and material shall flow steadily for continuous operation of the crusher. All portions of hoppers and chutes subject to abrasive wear caused by movement of the aggregate shall be lined or faced with replaceable abrasion-resistant steel plates not less than 1/4-inch thick conforming to ASTM A242. All bolt heads subjected to the flow of material shall not extend beyond surfaces exposed to the material and shall be of hardened steel.

3.9 Engine. The engine, including all systems, components, accessories and auxiliaries, shall conform to MIL-E-11275, group II, except as otherwise specified herein.

3.9.1 Power and speed rating. The power and speed rating of the engine shall be such that operation of the crusher under any of the operational requirements and conditions specified herein will not require horsepower in excess of the continuous horsepower rating of the engine for the applicable governed speed, as established by the Qualified Products List, QPL-11275.

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3.9.2 Fuel tank. The capacity of the tank shall be sufficient for 8 hours of continuous operation under any operating condition specified herein. Fuel capacity shall be based upon 85 percent of the fuel consumption rate obtained during conduct of Test Method 2000, MIL-STD-1400, at the applicable horsepower and governed speed.

3.9.3 Cranking system. An electric cranking system shall be furnished.

3.9.4 Charging system. The charging system shall conform to MIL-E-11275, and shall be 18 ampere capacity.

3.9.5 Batteries. Storage batteries conforming to MS35000 shall be furnished in the quantity required to meet all performance requirements specified herein.

3.9.6 Battery box. A battery box in accordance with MIL-E-11275 shall be furnished.

3.9.7 Control system. The control system shall include all controls specified in MIL-E-11275, and shall be of the engine shut-off type.

3.9.8 Housing. A steel housing shall be provided as specified in MIL-E-11275.

3.9.9 Instrumentation. Instruments shall include all instruments specified in MIL-E-11275. The instrument panel and its components shall be visible from and accessible to the operator's platform.

3.9.10 Air cleaner. The air cleaner shall be of the dry type, conforming to MIL-A-52363, class III. A rain cap shall be furnished. The air cleaner shall be provided with a restriction indicator conforming to MS53063-3.

3.9.11 Cooling system. A complete cooling system in accordance with MIL-E-11275 shall be furnished. The radiator shall conform to MS35773.

3.9.12 Clutch and clutch controls. The engine shall be equipped with an overcenter clutch which shall be accessible for lubrication, adjustment, and the replacement of parts. The clutch shall withstand repeated engagement without damage under any speed and load conditions specified herein. The lining shall conform to KKK-L-370, type II, and shall be installed in accordance with the lining manufacturer's instructions. Clutch control shall be accessible from the operator's platform. Clutch capacity shall be not less than 150 percent of maximum engine torque.

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3.10 Drives.

3.10.1 Chain drives. Unless otherwise specified herein, chains used in drive trains shall be in accordance with MIL-C-52058 and shall conform to MIL-STD-421, MIL-STD-423, or MIL-STD-424, as applicable. Chain drives shall be provided with idlers or other means of adjustment.

3.10.2 V-belt drives. V-belt drives shall be in accordance with ASA B55.1; or Rubber Manufacturers Association, Inc., "Engineering Standard for Drives Using Narrow V-Belts (3V, 5V, and 8V Cross Sections)." Jack shaft drives may be used. All V-belt drives shall have provision for adjustment of belt tension.

3.10.3 Gear drives. Gear drives shall conform to ASA B6.6, class 2 or class 3. Gear drives shall be totally enclosed and shall run in an oil bath. The enclosures shall be as specified in 3.17.4. Cast iron gears when used shall conform to ASTM A536. The analysis and the heat treatment shall be as recommended by the supplier. When cast iron gears are provided, the pinion shall be of steel.

3.10.4 Shafting. Shafting 2 inches in diameter and larger shall be of steel in accordance with ASTM A193, type B7, Rockwell hardness 25 to 32 on Rockwell "C" scale; and shall be turned, ground and polished. The fit tolerance between all shafting and driven members shall not exceed 0.003 inch.

3.10.4.1 Keys and keyways. All keys and keyways, except for the engine shall be machined in accordance with figure 1. Key ends shall be rounded to prevent line or point contact after installation. Keys shall be at least as hard as the shafting.

3.11 Feeder. The material shall be fed to the crusher by means of a chute-type feeder. The loading height shall be not more than 92 inches. The top opening of the hopper shall be rectangular. The top opening dimensions of the hopper shall be sufficient to extend not less than 6 inches (measured horizontally) from the front and sides of the crusher opening, and rearward a sufficient distance so that material dropping vertically from any point over the rear edge of the hopper will not fall on any part of the crusher. The sides of the hopper shall slope downward from the top opening so as to discharge into the crusher opening. The hopper shall be constructed of steel plate having a nominal thickness of not less than 0.250 inch. All interior surfaces shall be faced with abrasion resistant steel plates as specified in 3.8. The hopper shall withstand the impact of maximum size rock falling on any part of the hopper while charging the crusher without permanent deformation or damage.

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3.12 Crushing unit. The crushing unit shall be of the jaw type with overhead eccentric action. The opening of the jaws measured at the top shall be not less than 14 by 24 inches at nominal 3-inch jaw opening. The discharge opening at closed stroke shall be adjustable over the range from not more than 1 inch to not less than 4 inches with one toggle plate. The crushing unit shall be powered by an integrally mounted gasoline engine as specified in 3.9. The crushing unit shall crush materials ranging in size from a cross section having length and width dimensions 2 inches less than the top opening dimensions of the crusher jaws down to the size of the minimum jaw setting. The maximum size rock for a 14 by 24 inch crusher shall be 12 by 22 inches.

3.12.1 Frame. The crushing unit frame shall be constructed of cast steel or welded steel plate and shall withstand any stress caused by operation as specified herein without exceeding the stress limitations specified in 3.1.2. The frame, if of welded steel plate, shall be stress relieved after welding in accordance with the Standard Qualification Procedure of the American Welding Society for weldments of this size. The frame, if of cast steel, shall be in accordance with QQ-S-681, class 2, 3, or 4. The contact points on the frame which support the parts that transmit the crushing stresses to the frame shall be machined, or provisions shall be made to distribute the crushing load on the frame so that the frame will withstand the stresses imposed when crushing maximum-size material.

3.12.2 Jaw and cheek plates. Jaw and cheek plates shall be made of manganese steel conforming to ASTM A128 with analysis and heat treatment as recommended by the supplier of the crusher. The movable and stationary jaw plates shall be reversible and shall be secured so that they will not loosen and shift during the crushing operation. The back surfaces of the jaw plates shall be in a plane so that they will fit against their respective supporting member and provide equal load pressures in operation. The jaw plates shall be of the heavy duty type and shall be not less than 2-1/2-inches thick at the main crushing area measured from the back bearing surface of the plate to the valley of the corrugations and shall be not less than 1-3/4-inches thick at the top and bottom of the plate measured in the same manner. Corrugations shall be of a general sinusoidal configuration and shall be not less than 3/4-inch deep measured from crest to valley. The jaw plates shall be removable for replacement without the removal of other accessories or components, except cheek plates and wedges. The jaw plates shall have inserts for attaching lifting eyes or similar means for removing or inserting new jaw plates. Cheek plates shall be mounted on the frame between both sides of the jaw plates to protect the frame during the crushing operation. The cheek plates shall be constructed of two or more pieces which shall be replaceable and shall not loosen or shift during the crushing operation.

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3.12.3 Pitman. The pitman shall be made of cast or fabricated steel reinforced with ribs and machines to receive the movable jaw plate. If the pitman is constructed of cast steel, the steel shall be in accordance with QQ-S-681, class 2, 3, or 4. If the pitman is of fabricated steel, the steel shall be stress relieved in accordance with the Standard Qualification Procedure of the American Welding Society for weldments of this size. At the lower, or toggle end of the pitman, provision shall be made to receive a renewable toggle seat.

3.12.4 Pitman shaft. The pitman shaft shall be an eccentric shaft machined from steel conforming to SAE 4140, 4340, or 6150, and shall be heat-treated to withstand operation as specified herein without exceeding the stress limitations specified in 3.1.2. The main bearings shall be mounted in bearing housings on the crusher frame.

3.12.5 Pitman shaft bearings. The pitman and pitman shaft shall be mounted on antifriction bearings. The bearings shall be protected by labyrinth-type grease and dust seals which shall prevent dust and foreign matter from entering the bearings. The bearings shall conform to FF-B-185 or FF-B-187, except that taper bore bearings and taper sleeve adapters will be permitted.

3.12.6 Flywheels. Two cast flywheels shall be mounted on the pitman shaft and shall be counterbalanced to offset the mass of the eccentric revolving parts and be of sufficient size to provide inertia to transmit and equalize the energy necessary for crushing maximum size rocks without varing the speed more than 10 percent. One flywheel shall be machined or grooved for V-belts and used as a driving pulley. The drive shall conform to 3.10.2.

3.12.7 Toggle plate. A cast iron or steel toggle plate shall be provided to rest between the toggle seat on the pitman and the seat in the toggle wedge of the adjusting mechanism. The toggle plate shall be held in place by means of a tension bolt and spring arranged so that the bolt will not bind or bend during operation of the crusher or during adjustment of the plate. The toggle plate shall be constructed so that it will break or deform when uncrushable material such as tramp iron enters the crushing unit and shall fail before any other part of the crusher is stressed beyond the limitations specified in 3.1.2. The toggle plate shall not break or permanently deform when crushing rock as specified in 3.1.1 and 3.12. The toggle plate shall be accessible for replacement without the removal of other components of the crusher.

3.12.8 Adjusting mechanism. An adjusting mechanism of the wedge-and-screw or shim type shall be provided which will permit adjustment of the opening between the jaws. The adjusting mechanism shall be of cast steel or fabricated steel plate and shall be reinforced to withstand crushing of maximum size material. All contact surfaces of the adjusting mechanism shall be machined to insure uniform distribution

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of the load on the respective parts of the crushing-unit frame. A stop shall be provided to prevent adjustment to actual contact of the jaws during the crushing cycle.

3.12.9 Jaw setting. The jaw setting shall be measured along or parallel to the longitudinal axis of the crushing unit on a horizontal plane. Measurement shall be of the greatest opening in the minimum discharge position of the jaws in the crushing cycle, normally between the ridge of one jaw and the valley of the other. Variation between any two measurements for any jaw setting shall be not greater than 1/4 inch.

3.13 Conveyor. The crusher shall be equipped with an end-delivery conveyor. The conveyor shall have a belt 18 inches in width and shall receive materials from the crushing unit. The centerline of the head pulley shall be not less than 8 feet, 4 inches from the ground. The discharge end of the conveyor shall be hinged for transport purposes, and a saddle with a tiedown device shall be provided to rigidly secure the free end of the conveyor when folded in the transport position. Normal belt speed shall be not more than 300 feet per minute.

3.13.1 Frame. The conveyor frame shall be of sufficient strength to support all other components of the conveyor not supported directly by the chassis. The frame shall support all dead and live loads imposed by other components when operating with a loading of 25 pounds per lineal foot of belt in the carrying run.

3.13.2 Conveyor drive pulley. The pulley shall be of steel construction, not less than 20 inches long, not less than 10 inches in diameter, crowned, and mounted on antifriction bearings. The crown shall taper 1/8 inch per foot to the center of the pulley. The outside surface of the pulleys shall be concentric with the shaft within a total indicator reading of 0.060 inch, except that a total indicator reading of 0.090 inch is permissible in an area immediately adjacent to a longitudinal weld. The pulley supplying power to the belt shall be lagged.

3.13.3 Tail pulley. Unless used as the drive pulley, the tail pulley shall be of steel construction, crowned, wing, or self-cleaning type, and mounted on antifriction bearings. The pulley shall be not less than 10 inches in diameter and not less than 20 inches long. The outside surface of the pulley shall be concentric with the shaft within a total indicator reading of 0.100 inch. When the tail pulley is used as the drive pulley, it shall conform to 3.13.2.

3.13.4 Taper-type locking device (pulley and shaft). All pulleys shall be provided with an inner hub which clamps to the shaft by moving axially along a taper in the outer hub. The mechanical clamping shall be accomplished by the use of a screw member or members providing for mounting and removal of the pulley.

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3.13.5 Bearings. Bearings for pulleys, takeups, and drive shall be heavy duty, antifriction type, conforming to FF-B-171, FF-B-185, or FF-B-187. Bearings conforming to FF-B-171 may be retained on the shaft by the bearing-retainer plate and shims between the retainer plate and housing to provide endwise adjustment or by means of a split-tapered sleeve. When bearings conforming to FF-B-185 are mounted in a manner other than being secured against shoulders, the application shall be subject to the approval of the contracting officer for the specific application.

3.13.6 Takeup adjustment. To sustain belt tension, a takeup adjustment consisting of two 1-inch Acme thread arrangements (one per side) shall be provided at the tail pulley. The total available belt takeup on each side shall be not less than 12 inches. Threads shall conform to ASA B1.5.

3.13.7 Belt carriers. The conveyor belt shall be carried between pulleys on troughing rolls, impact rolls, and return rolls.

3.13.7.1 Troughing rolls. The troughing rolls shall be not less than 4 inches in diameter. Rolls and roll assembly shall conform to MS16961; however, dimensions shall be modified to make the carrier compatible with the belt specified in 3.13.8. The assemblies shall be mounted at intervals of not more than 4 feet.

3.13.7.2 Impact rolls. The impact rolls shall be not less than 4 inches in diameter. Rolls and roll assembly shall conform to MS16963; however, dimensions shall be modified to make the carrier compatible with the belt specified in 3.13.8. The assemblies shall be mounted at intervals of not more than 1-1/2 feet and shall support the belt under the receiving hopper and whenever rock falls from the discharge of the crushing unit.

3.13.7.3 Return rolls. The return rools shall be not less than 4 inches in diameter. Rolls and roll assembly shall conform to MS16964; however, dimensions shall be modified to make the return roll compatible with the belt specified in 3.13.8. The assemblies shall be mounted at intervals of not more than 9 feet.

3.13.8 Conveyor belt. The conveyor belt shall conform to ZZ-B-206, type A, grade 375, 4 ply, 18 inches in width, and with the following exceptions: The top cover shall be 1/8 inch thick, and the bottom cover shall be 1/16 inch thick with a skim coat between all plies; the tensile strength of the top and bottom covers shall be not less than 2,000 psi; and the friction pull shall be not less than 15 pounds. The belt fabric shall be treated to resist the growth of fungus.

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3.13.8.1 Belt fasteners. The belt shall be furnished with bolted metal plate fasteners to provide a belt joint for assembly and dissassembly of the belt.

3.13.9 Receiving hopper. The receiving hopper shall be mounted at the lower end of the conveyor frame and shall be removable. The hopper shall be of the skirt-board type and shall have three sloping sides to prevent spillage of material. The hopper shall extend not less than 12 inches beyond the discharge of the crusher jaws toward the discharge end of the conveyor. The space between the three sides of the hopper and the conveyor belt shall be provided with adjustable and replaceable rubber flashing mounted under the hopper plates. The hopper plates shall be as specified in 3.8. The flashing shall be 1/4-inch thick, 5 percent wax impregnated, abrasion resistant, natural rubber compound of 2,500 to 3,000 psi tensile strength, 500 percent elongation 40 to 50 durometer hardness.

3.13.10 Drive mechanism. Power for the conveyor shall be supplied from the gasoline engine specified in 3.9 by a drive as specified in 3.10 and may include a reducing gear. The drive shall be arranged so that the conveyor will run at all times when the crusher flywheel is turning. A takeup, or takeups, shall be furnished for installation and for maintaining tension in V-belts or chains. If furnished, reducing gears shall be totally enclosed and shall operate in an oil bath. Gears shall conform to ASA B6.6, class II. Oil temperature in bearings and gear cases shall not rise more than 100° F. above ambient temperature.

3.13.11 Belt cleaner. A weighted or spring-loaded, plow-type, rubber-blade scraper shall be mounted inside the return run of the belt at the foot (tail) end of the conveyor. The scraper shall exert uniform pressure across the full width of the belt. The blades shall be neoprene rubber and shall be not less than 3/8-inch thick. The tensile strength of the blades shall be not less than 1,200 psi and the durometer hardness shall be 65 to 75.

3.14 Platforms and walkways. Platforms and walkways constructed of checkered plate, steel grating, or expanded metal shall be furnished to provide access to all points requiring routine maintenance, adjustment, or operational control. The platforms and walkways adjacent to the feeder hopper shall withstand the impact of maximum size rock falling onto the platform and walkways while charging the crusher without permanent deformation or damage.

3.14.1 Operator's platform and walkways. An operator's platform not less than 3 feet wide shall extend along the front side of the feeder hopper. Walkways shall be provided on both sides of the feeder hopper and shall extend from the rear side of the feeder hopper to

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the front side of the operator's platform. Walkways shall be not less than 18 inches wide but of sufficient width so that material dropping vertically over the outside edges will not fall on any part of the crusher. The operator's platform and each walkway shall be rectangular and all shall lie in the same horizontal plane. The operator's platform and walkways shall be at the same elevation, plus zero minus one inch, as the top rim of the feeder hopper. A ladder providing access to the operator's platform shall be furnished as an integral part of the crusher. Removable handrails shall be provided on the outer sides of the operator's platform and walkways and at the ends of the walkways. Handrails shall be continuous, except at the access ladder.

3.14.2 Engine-servicing platforms. A platform shall be provided on each side of the engine from which the operator may service and operate the engine. Provisions shall be made for holding and securing the platforms for transport purposes, if required to meet other requirements of this specification. The engine-servicing platforms shall be at an elevation to permit access to the engine. Each platform shall lie in a horizontal plane.

3.15 Trailer components. The crusher shall be of the full trailer type with wagon steer towbar and front axle assembly. The crusher trailer components shall be compatible for towing with a Military 5 ton, 6 x 6, M51 dump truck conforming to MS500010.

3.15.1 Towbar and front axle assembly. A lunette conforming to MS51337 shall be inserted in a support which shall be welded into a hinged towbar The fit of the support shall conform to ASA B4.1, class LC 10. The lunette shall be supported the entire length of the shank. When the assembly is in towing position, the height of the lunette shall be between 30 and 34 inches from the ground. The center of the lunette and the centerline of the towbar shall be located on the longitudinal centerline of the crusher within a 1/8 inch tolerance either side. The towbar shall be furnished with a vertically adjustable landing leg so that coupling or removing the lunette from the towing pintle can be accomplished by one man. The landing leg shall be of of the crank operated type and shall either retract or be hinged to fold into the travel position to provide not less than 22 inches ground clearance. The leg shall provide positive locking in the parked and in the travel positions. The mechanism shall be protected from exposure to dirt and road splash. The leg shall be equipped with a mud pad not less than 6 inches in diameter.

3.15.1.1 Safety chains. Two safety chains or cables with hooks shall be furnished and shall conform to the ICC regulations as specified in 3.5 for full trailers. The safety chains shall be installed on the crusher so that they extend between 36 to 42 inches toward the truck from the center of the lunette eye. The hooks shall be of the safety type that may be wired closed and shall be compatible with the safety chain shackles of the Military truck specified in 3.15.

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3.15.1.2 Intervehicular connections. The crusher shall be furnished with an intervehicular electrical harness and two air brake lines for connecting the truck electrical and air brake systems to the crusher systems. The air brake service line shall be located on the curb side and the emergency line shall be on the road side. All intervehicular connections shall be installed on and fastened to the crusher towbar so that they extend between 36 and 42 inches toward the truck from the center of the lunette eye. From the towbar, these intervehicular connections shall be rounted to the trailer connections on the front of the crusher located in accordance with SAE J702 and prevented from entanglement when the towbar and front axle are jack-knifed up to an angle of 110 degrees to the right and to the left while coupled to the truck. Electrical harness plug clips and air brake dummy couplings shall be provided to hold the intervehicular components doubled back and clear of the ground when the crusher is prepared for crushing rock and for storage. Electrical and air brake components shall be as specified in 3.15.6 and 3.15.5.1, respectively.

3.15.1.3 Fifth wheel suspension assembly. The crusher shall be equipped with a sprung, fifth wheel suspension assembly supporting the front of the crusher. The fifth wheel shall be of the rigid mounted, fixed type having a lubricated thrust plate between the upper and lower bearing surfaces. The thrust plate shall be not less than 20 inches in diameter. The fifth wheel design shall eliminate play and provide automatic compensation for wear. The center of the fifth wheel shall be located on the longitudinal centerline of the crusher within 1/8 inch tolerance to either side. The suspension shall provide the articulation specified in 3.7.2 and shall maintain axle alinement with respect to the crusher. The suspension shall be attached to the axle at the spring centers specified for the axle. U-bolts or twin studs shall be equipped with self-locking nuts of not smaller than 7/8 - 14 size.

3.15.1.4 Front axle. The front axle shall conform to MS35865-A-4, except that the foundation brakes shall be of the straight air type. The portion of the crusher GVW imposed on the front axle shall not exceed 10,000 pounds.

3.15.2 Frame. The crusher framing members shall be constructed to withstand all operation and travel forces as specified herein without permanent deformation or damage.

3.15.3 Rear axle. The rear axle shall conform to MS35865-A-4, except that the foundation brakes shall be of the straight air type. The crusher frame shall be attached to the axle at the spring centers specified for the axle. Attaching U-bolts or twin studs shall be equipped with self-locking nuts of not smaller than 7/8 - 14 size. The portion of the crusher GVW imposed on the rear axle shall be

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between 50 and 65 percent of the GVW and shall not exceed 10,000 pounds. The rear axle shall be installed perpendicular to and symmetrical about the crusher's longitudinal centerline. The difference between the two diagonal measurements from the true center of the fixed fifth wheel on the crusher centerline to the center end of each rear axle spindle shall not exceed 1/8 inch.

3.15.4 Wheel and tire assemblies. The crusher shall travel on four, single wheels. Four tires, tubes, flaps, and wheels shall be furnished. Tires and wheels shall conform to MS35388-21 and MS53044-5 respectively for the axles specified herein. The 11:00 by 20 inner tubes shall be in accordance with MS35392-15 and shall conform to ZZ-I-550. The load imposed on each tire, measured at the ground, shall not exceed the maximum rated load specified in MS35388. Each assembly of tire, tube, flap, and wheel shall be balanced within industry limits to preclude wheel tramp or drumming at road speeds up to 45 mph.

3.15.5 Brake system. The crusher shall be equipped with a waterproof service and emergency breakaway air brake system actuating foundation brakes on all wheels. The brake system shall comply with the ICC requirements in 3.5, except as specified herein. The brake system shall include service and emergency intervehicular lines as specified in 3.15.1.2, service and emergency line filters, an SAE J10 air reservoir, an emergency relay valve in accordance with MS53004, and air-actuated foundation brakes. The reservoir shall have an air volume capacity of not less than 1-1/2-times that recommended by SAE J813 for truck trailers and shall be equipped with an end drain cock. The foundation brakes shall be of the right angle, self-adjusting type and shall be correctly balanced in application pressure and timing to reduce possibility of jack-knifing the crusher during stops. The rear axle brake shall have greater effectiveness when applied than the front axle brakes through the use of smaller chambers on the front than on the rear or by installation of a pressure holding or reducing valve in the front axle line. All brake system components and line shall be located, installed, and protected to prevent damage or malfunction from ground obstructions and debris, from falling rock when operating, and to obtain maximum efficiency and ease of maintenance.

3.15.5.1 Brakelines. The intervehicle air lines shall be furnished with coil spring hose guards, identification tags, and connectors (gladhands) having metal spring dust covers on each end of the lines. The brake lines shall be routed within the protection of the chassis framing and shall be secured by coated tubing clamps spaced at intervals not to exceed 3 feet. Grommets or through-frame connectors shall be installed where lines pass through sheet metal and framing members. The lines connecting to the emergency relay valve shall be of flexible hose for a minimum of 10 inches prior to the fitting, and sufficient slack shall be provided to permit interchangeability of valves made by more than one supplier. Brake-line components shall conform to the following:

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(a) Flexible air hose - MIL-H-3992, type I.
(b) Rigid air line - SAE J844, copper tubing.
(c) Fittings - SAE J512, heavy duty, air-brake, compression type, reuseable.
(d) Intervehicular couplings - MS35746, and furnished with spring-loaded covers.
(e) Identification tags - MS53007 near each gladhand.

3.15.5.2 Brake lining. The brake lining material shall contain a sintered, iron-based lining mixture which shall be fused to a steel backing and installed on the brake shoes by the use of steel bolts or rivets following the lining manufacturer's recommendations. The lining segment thickness, including the steel backing, shall be not less than 3/8 inch and shall be radially ground to fit the brake drum. The lining mixture shall conform to KKK-L-370, type II, except that the kinetic coefficient of friction at surface speeds of 4,000 to 6,000 feet per minute shall exceed 0.30 and other characteristics of the material shall be as recommended by the lining manufacturer for airbrake usage.

3.15.5.3 Brake preservation. All metal surfaces enclosed by the brake drum including the interior surface of the drum, metal surfaces of the shoes, backing plates, and all attached parts shall be coated with a lacquer-resisting primer conforming to TT-P-664 or with a black oxide primer. Wedges, pins, eccentrics, cam levers, anchors, and retracting springs shall be coated with preservative oil conforming to MIL-L-21260, grade 2. Items or surfaces which are wholly prime coated, painted in accordance with Government specifications, or constructed of materials providing corrosion protection equivalent to that provided by MIL-C-16173, grade 1, do not require additional protection. The working surfaces of the brake linings shall not be painted or coated.

3.15.5.4 Chock blocks and cradles. Four wood chock blocks with captive chains and four wire cradles shall be furnished. The blocks and cradles shall be installed, two for the front wheels and two for the rear wheels. The hook on each captive chain shall be snapped on the outer loop of each cradle to allow lifting the chock, including its coiled chain, from its cradle and placement of the chock both ahead of and behind its respective wheel. The chock blocks shall conform to figure 2 and the cradles shall conform to figure 3.

3.15.6 Vehicular electrical system. The crusher shall be equipped with a waterproof, 24-volt, vehicular electrical system for service and blackout illumination during travel. The service electrical system shall be in accordance with the SAE Handbook, the American Trucking Association (ATA) specification E-3-1961, and ICC Motor Carrier Safety Regulations, except as specified herein. The blackout electrical system shall be as specified herein, using the wiring diagram shown on Drawing 8347201 for guidance. All wiring and components shall be located and guarded to provide protection from falling and flying rock and other damage, breakage, and vibration. The electrical system shall be in accordance with table I. Item quantities that are not specified shall be furnished as required (AR) for each crusher.

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Table I. Vehicle electrical system

Quantity	Description	Requirements
AR	Service light and reflector location.	ICC Regualtions and ATA Spec. E-3-1961.
14 ea. min.	Service lights; clearance, side-marker and identification.	MS35423.
8 ea.	Blackout lights; clearance, side-marker.	MS35424.
2 ea.	Service combination lights.	SAE-SIT 4-in-dia, water-proof; SAE J759.
2 ea.	Blackout combination lights.	MS51330.
8 ea.	Reflectors.	MIL-R-12144.
1 ea.	Intervehicular harness assembly.	2 ea. MS75020 plugs. cable MIL-P-13486/12, length as specified herein.
1 ea.	Front trailer receptacle.	MS75021-1, with cover.
ea.	Rear trailer receptacle.	MS75021-2 with cover.
AR	Low-tension cable.	MIL-C-13486, type I, class as applicable.
AR	Harness insulation.	MIL-I-631 and MIL-I-15126.
AR	Loom.	SAE J562, oil proof.
AR	Cable support straps.	MS21322, MS21333, and MS21334.
AR	Circuit number bands.	MS39020.
AR	Friction retainment connectors.	MS27142 through MS27148, and MS33800.
AR	Connector lubricant.	MIL-S-8660.
AR	Connector holder clip.	Spring metal or plastic type, one for each friction connector.
AR	Terminal connectors.	SAE J561, spade with insulation grips.
3 ea.	Circuit breakers.	MS39062.

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3.15.6.1 Lighting. The crusher shall be equipped with both service and blackout lighting. The service lighting shall be as specified in table I and the blackout lighting shall be as specified herein using the wiring diagram as a guide. The lights shall be provided with shock and vibration protection by location and through use of resilient mountings to prolong filament life. Identification, side-marker, and clearance lights shall be clustered and mounted on plates inserted into lamp bracket sockets conforming to SAE J725. These clusters shall be removable for storage and identified with plates (see 3.21) to preclude incorrect replacement. Side-marker and clearance lights, both service and blackout, shall be mounted on the same mounting at each corner. The rear service combination lights shall be of the flush mount, waterproof type having either a sealed unit mounted in an elastomer ring, or a bulb and lens mounted in an elastomer housing. These lights shall be installed as far apart as practicable to indicate crusher width and for turn-signal operation. The blackout combination lights shall be installed toward the inside or below the service combination lights. These four lights shall be protected by recessing the lights not less than 1/2 inch nor more than 2 inches within rings of 6-inch minimum diameter standard weight pipe.

3.15.6.2 Wiring. The wiring shall be fabricated and installed in accordance with the "Electrical Equipment and Lighting" section of the SAE Handbook, and the wiring diagram, except as specified herein. Cable clamps, grommets, loom, and commercial thin-wall conduit shall be used to prevent whipping, chafing, or other mechanical damage to the wires. Thin-wall conduit shall be provided where wires are not routed within the protection of framing members. Splices shall be limited to junctions within the harnesses and shall be waterproofed with insulation. Wires adjacent to friction connectors and junctions and two or more wires following the same routing and not protected by conduit shall be formed into a harness with either 0.010-inch-thick, 1-inch-wide tape or insulation. All wires shall be number coded near each connection following the wiring diagram and as specified herein. Friction connectors shall be lubricated with silicon connector lubricant prior to connection, and each assembly shall be secured to the crusher with a connector holder clip. A splashproof, electrical, semitrailer-type nose-box shall be furnished to contain terminal connectors for all live circuits and the two ground circuits and for three service lighting circuit breakers. The circuit breakers shall be wired to protect circuits 22-460, 22-461, and 21-489. Each cluster light assembly shall be provided with two or three pigtail leads with friction connectors to provide live service and blackout or service only leads as applicable with ground return. The service combination lights shall be connected through pigtail leads with friction connectors to circuits 21, tail, and 22-460 or 22-461, stop and turn. The rear blackout combination lights shall be connected to circuits 24, tail, and 23, stop. The rear receptacle shall be wired into the system to provide lighting of other towed vehicles.

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3.15.6.3 Reflectors. Reflectors shall be furnished and mounted as required by ICC Motor Carrier Safety Regulations. In addition to ICC requirements, two amber reflectors shall be mounted on the front of the crusher as far apart as practicable and at the same height as those mounted on the sides.

3.15.7 Stabilizing jacks. The crusher shall be equipped with not less than four screw-action stabilizing jacks permanently attached to the chassis. The jacks shall conform to GGG-J-51, type I, class 3, rated capacity 8-ton. The jack shall have a steel ball bearing cap. The base of the jack shall be fully enclosed. Detachable pads may be furnished. Provisions shall be made for stowing the pads on the trailer in an identified, accessible location.

3.16 Electromagnetic compatibility. Unless otherwise specified herein the electromagnetic compatibility of the crusher shall be in accordance with MIL-E-55301, tactical support class. When specified (see 6.2), the electromagnetic compatibility of the crusher shall be in accordance with MIL-E-55301, administrative class.

3.17 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.17.1 Lubricants. The crusher shall operate as specified herein when lubricated with Military lubricants (see 6.3).

3.17.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II, or III. Fittings shall be located in a protected position. Fittings shall be accessible to a grease gun conforming to MIL-G-3859, equipped with a straight hydraulic coupler and a flexible extension conforming to MIL-L-4387, type III, class 2. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Engine-housing side panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

3.17.3 Pressure-release device. An automatic pressure-release device shall be provided where pressure-lubricating equipment could damage grease seals or other parts.

3.17.4 Filling, draining, and checking provisions. Enclosures such as gear cases and transmission housings that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with lubricant and for draining. The drain shall be fitted with a magnetic drain plug, size conforming to MS35844 or MS49006. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the crusher is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to the ground.

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3.17.5 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases conforming to MIL-G-10924 or MIL-G-23827, as applicable. The crusher shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because Military greases are not compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which Military grease has been used.

3.17.6 Oil lubrication. Unless otherwise preserved for packaging, all components shall be filled to the operating level with winter grades of Military oil designated for use in the temperature range of 0° F. to 32° F. Each component shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricants used.

3.18 Fungus and moisture resistance. Unless otherwise specified (see 6.2), the electrical circuitry, including all components and connections, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173 with the following exceptions:

(a) Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.19 Lifting attachments. Permanently-affixed attachments that enable the crusher to be lifted in its normal travel position shall be provided. Each attachment shall withstand, without damage to any part of the crusher, 2-1/2 times the resultant vectorial force imposed when the crusher is suspended by the lifting attachments. Attachments for multiple slings shall be located so that (a) slings will converge at not more than 19 feet above the lowest extremity of the crusher, and (b) not less than 1-inch clearance will be maintained between the slings and the crusher. The inside diameter of the attachment eye shall be not less than 3 inches. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.20.

3.20 Tiedown attachments. Permanently-affixed attachments that permit fastening the crusher to the floor or deck of a transportation medium shall be provided. Each attachment shall withstand, without damage to the crusher

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its proportion of the following static loads applied to the crusher: 8G forward, 8G rearward, 2-1/4G upward, 1-1/2G sideward, where G equals cargo weight of the crusher. Attachments shall be rounded and beveled to prevent cutting of the tiedown material. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.19.

3.21 Identification marking. The crusher shall be identified in accordance with MIL-STD-130. The marking shall be applied to the crusher on plates conforming to MIL-P-514, type I, style 2, photographic plate.

3.22 Transportation data plates. Transportation data plates shall be furnished and shall conform to MIL-P-514, type V, photographic plate. Location of lifting and tiedown attachments and recommended lifting and tiedown arrangements shall be included on these plates.

3.23 Instruction plates. Operating instruction plates conforming to MIL-P-514, type IX, photographic plate, shall be provided and shall indicate sequence for start up, shut down, and normal operation. Warning or caution plates conforming to MIL-P-514, type IX, photographic plate, shall be provided for all inherent dangers which can stem from unsafe operation or an unsafe operating position. The words "Warning" or "Caution" shall be red. The shade of red shall conform to FED. STD. No. 595, No. 31136.

3.23.1 Production capacity plate. A production capacity plate conforming to figure 4 shall be provided. Capacity figures for the crusher shall be included on the plate. Material shall be identical to that specified for the identification plate.

3.24 Treatment and painting. The portions of the crusher normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.25 Vehicle marking. The marking of the crusher shall be in accordance with MIL-STD-642. The gross vehicle weight of the crusher shall be marked on each side in such a manner as to be discernible. Lifting and tiedown attachments shall be marked "LIFT HERE" and "TIE DOWN HERE," as applicable. The prescribed tire pressure shall be marked near the tires.

3.26 Technical publications. Such technical publications as are specified shall be furnished (see 6.2).

3.27 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished (see 6.2).

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3.28 Rifle case. One rifle case with holder (FSN 2590-505-6736) will be furnished by the Government for each crusher (see 6.4). The rifle case shall be mounted on the crusher by the supplier in a location accessible to the operator. Mounting fasteners and fittings used shall be furnished by the supplier. The proposed location and method of mounting shall be subject to the approval of the contracting officer prior to installation.

3.29 Toolbox. The crusher shall be provided with a weatherproof toolbox made of steel not less than 0.071 inch thick. The toolbox lid shall be mounted on hinges conforming to MS35829 with a 0.125-inch-diameter brass pin. The toolbox lid shall open to at least 90 degrees from the closed position. The lid shall remain in the fully open position either by its own weight or by holding devices. Two hook-type tension fasteners shall be provided to keep the lid closed. Not less than a 3/4-inch clearance shall be provided between the hook lever and any other part of the toolbox. The fasteners shall be attached to the toolbox by either bolts or rivets. A hasp which will accommodate a lock conforming to MS35647 shall be provided for locking the toolbox. The toolbox shall be fastened to the crusher in a protected position. A drain hole with rubber plug shall be provided in the bottom of the toolbox. The toolbox shall be of a size to hold the tools specified (see 3.27).

3.30 Items to be furnished with the crusher. No shipment of crushers shall be made unless repair parts, tools, and accessories are included with the shipment or unless approval for shipment of the crusher without such items has been received from the contracting officer.

3.31 Workmanship. The crusher shall be free from defects such as incomplete welds, rust, cracks and other defects that could impair the operation of the crusher. Metal used in the fabrication of the crusher shall be free of kinks and sharp bends. Shearing and chipping shall be done neatly and accurately. Flame cutting of steel may be employed. Burned surfaces of flame-cut steel shall be ground or machined to remove all ash, burrs and rough edges. All bends shall be made with metal dies or fixtures to insure uniformity in size and shape. All threaded parts shall be in accordance with Handbook H28 unless otherwise specified herein. Rivets shall fill the holes completely. Rivetheads shall be full, neatly made, and in full contact with the surface of the member and shall be in accordance with SAE J492.

3.31.1 Welders and welding.

3.31.1.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting

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officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society.
Welding Qualifications of the ASME.

3.31.1.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are tested as specified herein.

3.31.2 Surfaces. All parts, components, and assemblies including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be cleaned and free of sand, dirt, fins, pits, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth, and all edges shall be rounded or beveled.

3.31.3 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein and on the applicable drawings. Unless prior approval is obtained from the contracting officer, all tapped holes shall be countersunk before tapping.

3.31.3.1 Line boring for main bearings on jaw crusher. Bearing support surfaces for each shaft shall be machined in a single setting to a tolerance of not more than 0.001 inch total indicator reading per foot of shaft. External bearing supports shall be doweled or shall be provided with an eccentric arrangement to assure reassembly in its initial position prior to final machining and installation of the bearings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

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4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The crusher shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The crusher shall be tested as specified in 4.6.2.1 through 4.6.2.6. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. An initial production crusher(s) will be selected at random by the Government from the first five crushers being produced from production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2.1 through 4.6.2.6 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production crusher shall not exclude the remaining crushers from the quality conformance inspection and acceptance provisions specified in section 4.

4.4.1 Inspection failure. Failure of an initial production crusher to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production crusher(s) and shall be cause for refusal by the Government to continue acceptance of crushers until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no coast to the Government on crushers previously accepted and produced under the contract. Any deficiencies found as a result of the initial production testing will be considered prima facie evidence that all crushers accepted prior to the completion of initial production testing are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each crusher shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each crusher shall be tested as specified in 4.6.2.7. Failure of the test shall be cause for rejection.

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4.6 Inspection procedure.

4.6.1 Examination. The crusher shall be examined for the following defects. (Items 116 through 120 shall be examined while the crusher is operating):

101. Material not as specified.
102. Parts or components missing or do not function.
103. Assembly incomplete or incorrect.
104. Safety devices or guards missing.
105. Workmanship not as specified.
106. Ease of maintenance not as specified.
107. Dimensions or weight not as specified.
108. Treatment and painting not as specified.
109. Identification or other marking, operating, and safety instructions, or manuals missing.
110. Lubrication devices not accessible.
111. Trailer components missing, inoperative, incorrect, or not as specified.
112. Brake preservation not as specified.
113. Discharge opening of crusher not adjustable over range as specified in 3.12.
114. Brake system not as specified or does not function.
115. Crusher engine and vehicular electrical systems not as specified, do not function, or are damaged. (Note: If crusher is examined using a 12-volt commercial truck (see 4.6.2.3.3), all lights should function but not necessarily at full brightness.)
116. Oil or grease seals leaking
117. Bearings or gear boxes overheating.
118. Shaft misalinement.
119. Belt carrier rolls do not turn.
120. Belt does not run straight on the centerline of the conveyor.

4.6.2 Tests.

4.6.2.1 Test conditions (operating). Prior to test, the crusher shall be lubricated with Military oils and greases as specified in 3.17.5 and 3.17.6. Oils shall be those designated for use in the ambient temperature at which the test is being performed. Quarry rock shall be as specified in 3.1.1.1.

4.6.2.1.1 Percentage of crush. A representative sample (not less than 50 tons) of crusher feed shall be taken and the percent of material larger than 1-1/2 inches shall be determined. Feed containing more than 20 percent of material smaller than 1-1/2 inches shall not be used.

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4.6.2.1.2 Operating time. Crushed rock production capacity shall be computed on the basis of net operating time which shall be defined as gross operating time less allowable downtime for the following:

(a) Jaming from rock fragments larger than the maximum size of rock specified in 3.12.
(b) Delay due to lack of material delivery to the power shovel or clamshell (see 4.6.2.1.3).
(c) Normal machinery maintenance (lubrication, jaw-setting adjustments, takeups).
(d) Other conditions considered by the contracting officer to be beyond the control of the supplier.

4.6.2.1.3 Loading. The crusher shall be loaded by a power shovel or clamshell.

4.6.2.2 Capacity and performance tests. Operate the crusher under the test conditions specified in 4.6.2.1 through 4.62.1.3 for 100 hours. Inability of the crusher to operate under the specified test conditions or as specified in 3.1.1 shall constitute failure of this test.

4.6.2.2.1 Engine performance. Prior to examination and tests of the crusher, the engine that will be installed in the preproduction model unit, shall be tested in accordance with Test Method Series 2000 of MIL-STD-1400. Nonconformance to 3.9.1 or failure to meet the requirements of Test Method Series 2000 of MIL-STD-1400 shall constitute failure of this test.

4.6.2.2.2 Flywheel test. Measure the speed of the flywheel during the capacity test for a continuous period of 3 minutes. Select a 3-minute period during which at least one large rock of maximum capacity size is being crushed. Nonconformance to 3.12.6 shall constitute failure of this test.

4.6.2.2.3 Uncrushable material test. With the crusher operating at capacity as specified in 3.1.1, drop a piece of uncrushable material, such as tramp iron, into the crusher. The piece shall be 1-1/2 plus or minus 1/2 inches larger than the maximum jaw opening at the jaw setting being used. Nonconformance to 3.12.7 shall constitute failure of this test.

4.6.2.3 Transporting test.

4.6.2.3.1 Turning test. With the crusher on level ground, turn the towbar from the centerline of the trailer to the right at least 110 degrees and to the left at least 110 degrees while keeping the lunette between 30 and 34 inches from the ground. Interference between members or nonconformance to 3.15.1.2 shall constitute failure of this test.

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4.6.2.3.2 Trailer articulation. Place diagonally opposite wheels of the trailer on platforms at least 8 inches high. Repeat the test for the other two wheels. Nonconformance to 3.7.2 shall constitute failure of this test.

4.6.2.3.3 Highway and cross-country test. Unless otherwise specified herein, tow the crusher over highways with a Military, 5-ton, 6 x 6, M51 dump truck which will be loaned by the Government (see 6.5). When specified (see 6.2), this test may be conducted using a commercial dump truck of not less than 32,000 pounds gross vehicle weight (GVW) rating and having full trailer towing components in accordance with SAE J849. Tow the crusher with full fuel tank over highways as specified in 3.7.1 for a distance of 100 miles at speeds up to 25 mph to sustain an average speed of not less than 20 mph. Using the same truck, tow the crusher over cross-country terrain as specified in 3.7.2 for a distance 50 miles at speeds up to 15 miles per hour. During the highway test bring the crusher to five complete, consecutive stops from a speed of 25 mph on a dry, level, smooth, paved surface. Immediately upon completion of the fifth stop accelerate to not less than 20 mph and bring the crusher to a sixth stop, measuring the stopping distance. Before any highway and cross-country tests are conducted, measure all critical mounting points on the crusher. After all highway and cross-country tests are completed, check the measurement of critical mounting points and magnaflux all welds. Damage to or permanent deformation of any part of the crusher, any change in critical mounting point dimensions, evidence of any weld being defective, incompatibility with the towing vehicle, any operating difficulty, or inability to stop the crusher within ICC requirements shall constitute failure of this test.

4.6.2.3.4 Trailer jackknife test. With the crusher in towing condition but not highway traveling, jackknife the trailer to the right and to the left as far as possible without damage to the crusher or truck. While towing the crusher, turn the truck to the right and to the left at its minimum turning radius. Nonconformance to 3.15 shall constitute failure of this test.

4.6.2.4 Lifting attachments. Lift the crusher using the lifting attachments. Measure the resultant vectorial force on each attachment during this lifting operation or calculate these forces. Then subject each attachment to a force 2-1/2 times greater than and in the same direction as the measured or calculated force. Any permanent deformation or breakage of lifting attachments, members to which attachments are fastened, fastenings, or any other part of the crusher, or sling convergence and clearance not as specified in 3.19 shall constitute failure of this test.

4.6.2.5 Tiedown attachments. Determine the amount and direction of resisting force on each attachment for each static load applied to the crusher in the amount and direction specified in 3.20. Then subject each

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attachment to each of these vectorial forces. If two or more attachments are identical and resist equal forces, only one of these attachments need be tested. Any permanent deformation or breakage of tiedown attachments, members to which attachments are fastened, fastenings or any other part of the crusher or cutting of the tiedown material shall constitute failure of this test.

4.6.2.6 Electromagnetic compatibility. The preproduction model, equipped for the reduction of radio interference in accordance with 3.16 will be tested as specified in MIL-E-55301. The supplier shall furnish the contracting officer the reports required by MIL-E-55301, except that the time interval for submission of reports shall be as specified (see 6.2). Upon approval of the reports by the contracting officer and provided all other requirements of the specification are met, the preproduction model shall be used as a model for all production crushers.

4.6.2.7 Production test. Lubricate each crusher as specified in 4.6.2.1 and operate for a period of 1 hour without crushing rock. Any operational difficulties, breakage, or permanent deformation shall constitute failure of this test.

4.7 Inspection comparison. The Government may select crushers at any time during the contract production period and subject these crushers to the examination specified in 4.6.1 and the tests specified in 4.6.2.1 through 4.6.2.6 to determine that the quality of the selected crusher is equal to the quality standards established during initial production testing. The inspection will be performed by the Government at a site selected by the Government. Crushers will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production crusher(s).

4.7.1 Inspection failure. Failure of any crusher when inspected as specified in 4.7 may be considered cause for refusal by the Government to continue acceptance of crushers until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.8 Inspection of preparation for delivery.

4.8.1 Preproduction pack inspection.

4.8.1.1 Examination. Examine the preproduction pack in accordance with 4.8.2.3. Presence of one or more defects shall be cause for rejection.

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4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completely processed crusher prepared for shipment shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. AQL shall be 2.5 percent defective.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects:

121.	Materials, methods, and containers not as specified for level A and B. Each incorrect material, method or container shall be considered one defect.	
122.	Bolts, nuts, pins, and washers removed not replaced in one of the mating parts and secured to prevent loss for level A and B.	
123.	Keys removed not secured as specified for level A and B.	
124.	Cloth matchmarking tags not waterproofed as specified for level A.	
125.	Surfaces of components requiring a contact preservative in accordance with the referenced document not coated for level A and B.	
126.	Trailer components not preserved and packaged in accordance with the referenced document for level A and B.	
127.	Engine, engine components and accessories not preserved in accordance with the referenced document.	
128.	Gear housings not filled with lubricant and gears actuated as specified for level A and B.	
129.	Drive chains not preserved as specified for level A and B.	
130.	Drive belts and pulleys not preserved as specified for level A and B.	
131.	Conveyor belt removed or not protected as specified for level A and B.	
132.	Repair parts not preserved in accordance with the referenced document.	
133.	Maintenance tools not preserved and packaged in accordance with the referenced document for level A and B.	
134.	Technical publications not preserved in accordance with the referenced document.	
135.	Intermediate packaging not as specified for level A and B.	
136.	Intermediate containers not secured to the crushers as specified for level A.	
137.	Toolbox lid not secured to prevent pilferage for level A.	
138.	Depreservation guide not prepared and attached.	
139.	Marking illegible, incorrect, or incomplete.	

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5. PREPARATION FOR DELIVERY

5.1 Preproduction pack. The preproduction pack is defined as a pack consisting of one completely processed crusher and all related boxed components which will be identical to the pack which the supplier will subsequently produce in fulfillment of the contract. Before packing a number of crushers, a preproduction pack shall be made and examined as specified for the applicable level to determine conformance to the preservation, packaging, packing, and marking requirements. Approval of the preproduction pack shall not relieve the supplier of his obligation to preserve, package, pack, and mark the crushers in conformance to this specification. Any changes or deviations of production packs from the approved preproduction pack shall be subject to the approval of the contracting officer.

5.2 Preservation and packaging. Preservation and packaging shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly and matchmarking.

5.2.1.1.1 Disassembly. Disassembly shall be the minimum necessary to safeguard parts subject to damage or loss and to accomplish reduction in cube. Bolts, nuts, pins, and washers removed shall be replaced in one of the mating parts and secured to prevent their loss. Keys removed shall be secured to the keyway of the primary component or shaft.

5.2.1.1.2 Matchmarking. When necessary to facilitate reassembly, detached components shall have matchmarking information on cloth tags conforming to UU-T-81, type A, or on metal tags. Cloth tags shall be waterproofed in accordance with MIL-STD-129. The tags shall be securely attached to the components so as to prevent damage to the preservation or to the components.

5.2.1.2 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.3 Unprotected metal surfaces. Unpainted surfaces of items such as hoppers, chutes, feeders, crusher frame, jaw and cheek plates, pitman, flywheels, toggle plates, adjusting mechanisms, rods, screws, clevises, nonprecision chains, conveyor rolls, exposed gears, sprockets, pulleys, ratchets, pins, linkages, shafts, drawbars, levers, rifle case, or any other unpainted metal surface requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

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5.2.1.4 Trailer components. Components of the trailer shall be preserved and packaged in accordance with the level A (mobile) requirements of MIL-STD-281.

5.2.1.5 Engine. The engine, engine components, and accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, as specified for type III classification.

5.2.1.6 Enclosed gears and gear housings. When the lubricant contained in the gear housing conforms to the specified operational lubricant, to MIL-L-2105, or to MIL-L-21260, a quantity of the applicable lubricant required to raise the lubricant to the operating level in the housing shall be added. When the lubricant does not conform to any of the lubricants specified herein, the gear housing shall be drained and refilled to the operating level with the applicable lubricant required for the operation. The gears shall be operated to insure coating all interior parts with the lubricant. The threads of the fill plug shall be coated with the lubricant before the plug is installed.

5.2.1.7 Drive chains.

5.2.1.7.1 Exposed chains. Exposed drive chains shall be coated with enough type P-3 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. After the excess preservative has drained, the chains shall be coated with type P-1 preservative. Sprockets shall be coated with type P-1 preservative.

5.2.1.7.2 Enclosed chains. Enclosed chains shall be preserved as specified in 5.2.1.6 for enclosed gears.

5.2.1.8 Drive belts and pulleys. Belts shall be removed or released from tension. Pulleys shall be coated with primer conforming to TT-P-664. Removed belts shall be preserved in accordance with MIL-P-116, method IC-1.

5.2.1.9 Conveyor and conveyor belt. The conveyor shall be folded as for transport. The belt shall not be removed but shall be protected from short radius bends, sharp corners of frame components or similar items.

5.2.1.10 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.2.1.11 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with the level A requirements of MIL-H-15424, and placed in the toolbox, if space permits. When sufficient space is not available, the tools shall be intermediate packaged as specified in 5.2.1.13.

5.2.1.12 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1.

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5.2.1.13 Intermediate packaging. Packaged and unpackaged components removed by disassembly, repair parts, technical publications, and maintenance tools that will not fit in the toolbox, shall be intermediate packaged in close-fitting boxes conforming to PPP-B-621, class 1, style optional or to PPP-B-601, domestic type, style optional. The contents shall be blocked and braced within the boxes to prevent free movement and damage to the contents. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. Strapping shall be zinc coated.

5.2.2 Level B. Each complete crusher shall be preserved and packaged as specified for level A with the following exceptions.

5.2.2.1 Matchmarking. Cloth matchmarking tags shall not be required to be waterproofed.

5.2.2.2 Exposed chains. Exposed drive chains shall be preserved as specified for level A, except only type P-3 preservative shall be applied.

5.2.2.3 Intermediate packaging. Boxes for intermediate packaging shall be domestic type or class as applicable but strapping is not required to be zinc coated.

5.2.3 Level C.

5.2.3.1 Engine. The engine, including the liquid cooling system, shall be preserved in accordance with the level C requirements of MIL-E-10062.

5.2.3.2 Repair parts, maintenance tools, and technical publications. Repair parts and technical publications shall be preserved and packaged as specified for level A. Maintenance tools shall not require preservation and packaging but shall be placed in the toolbox.

5.2.3.3 Other components. Preservation and packaging of other components of the crusher not specifically mentioned in 5.2.3.1 and 5.2.3.2 shall be limited to that protection deemed necessary by the supplier to provide protection against deterioration and damage from the supplier to the initial destination.

5.3 Packing. Packing shall be level A or C as specified.

5.3.1 Level A. The trailer mounted crusher shall be prepared for shipment unboxed (mobile). The toolbox lid shall be secured in a manner to prevent pilferage. The intermediate containers shall be placed on the crusher in any available space that will not increase cubage or interfere with lifting or towing and shall be secured in place with zinc coated strapping conforming to QQ-S-781 or QQ-S-790.

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5.3.2 Level C. The crusher, components, repair parts, maintenance tools, and technical publications shall be prepared for shipment unboxed (mobile) in compliance with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carrier rules applicable to the mode of transportation.

5.4 Depreservation guide. A depreservation guide shall be prepared, placed in a waterproof envelope marked "Depreservation Guide" and secured in a conspicuous location on the crusher. Unless otherwise specified (see 6.2 and 6.7) DA Form 2258 - Depreservation Guide for Engineer Equipment shall be used.

5.5 Marking. Packages, shipping containers, and unboxed (mobile) crushers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The crusher is intended for use in crushing and loading or piling quarry rock in the production of construction aggregate.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When an initial production crusher is not required (see 3.3).
(c) Number of crushers required for initial production inspection (see 3.3).
(d) When radio interference reduction in accordance with the administrative requirements of MIL-E-55301 is required (see 3.16).
(e) When fungus and moisture resistance other than that specified in 3.18 is required.
(f) Technical publications required (see 3.26).
(g) Repair parts and maintenance tools required (see 3.27).
(h) When commercial dump truck may be used for inspection (see 4.6.2.3.3).
(i) Time interval for submission of electromagnetic compatibility reports (see 4.6.2.6 and 6.6).
(j) Level of preservation and packaging and level of packing required (see 5.2 and 5.3). Level B preservation and packaging is intended to provide economical but limited protection and should be specified only when it is determined the crusher will be held in covered storage.
(k) When other than DA Form 2258 is to be used (see 5.4).

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6.3 Lubricants. The contracting officer should furnish a list of Military lubricants applicable to the crusher covered by this specification as contained in the Federal supply catalogue, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.4 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.28.

6.5 Government-loaned property. The contracting officer should arrange to loan the property listed in 4.6.2.3.3.

6.6 Electromagnetic compatibility. The contracting officer should submit a copy of the report (see 4.6.2.6) to the Commanding General, U. S. Army Electronics Command, ATTN: AMSEL-RD-GF, Fort Monmouth, New Jersey.

6.7 Depreservation guide. The contracting officer should arrange to furnish necessary copies of DA Form 2258, when applicable.

6.8 Revision changes. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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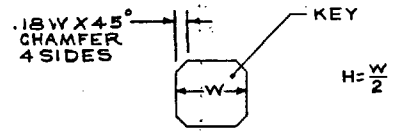
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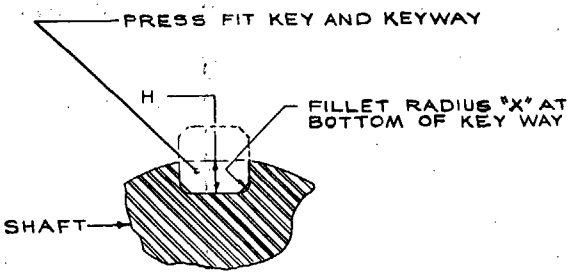
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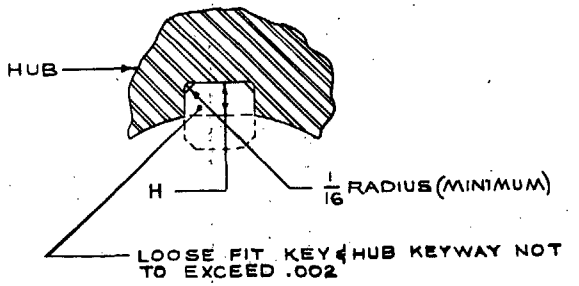
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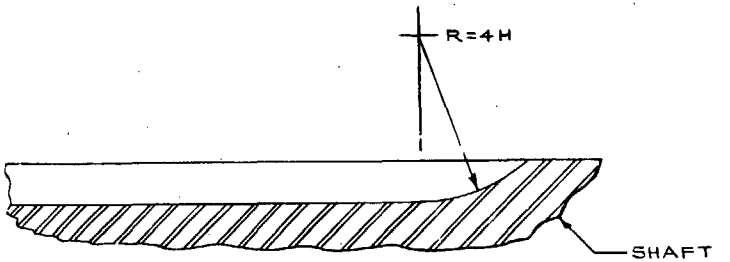
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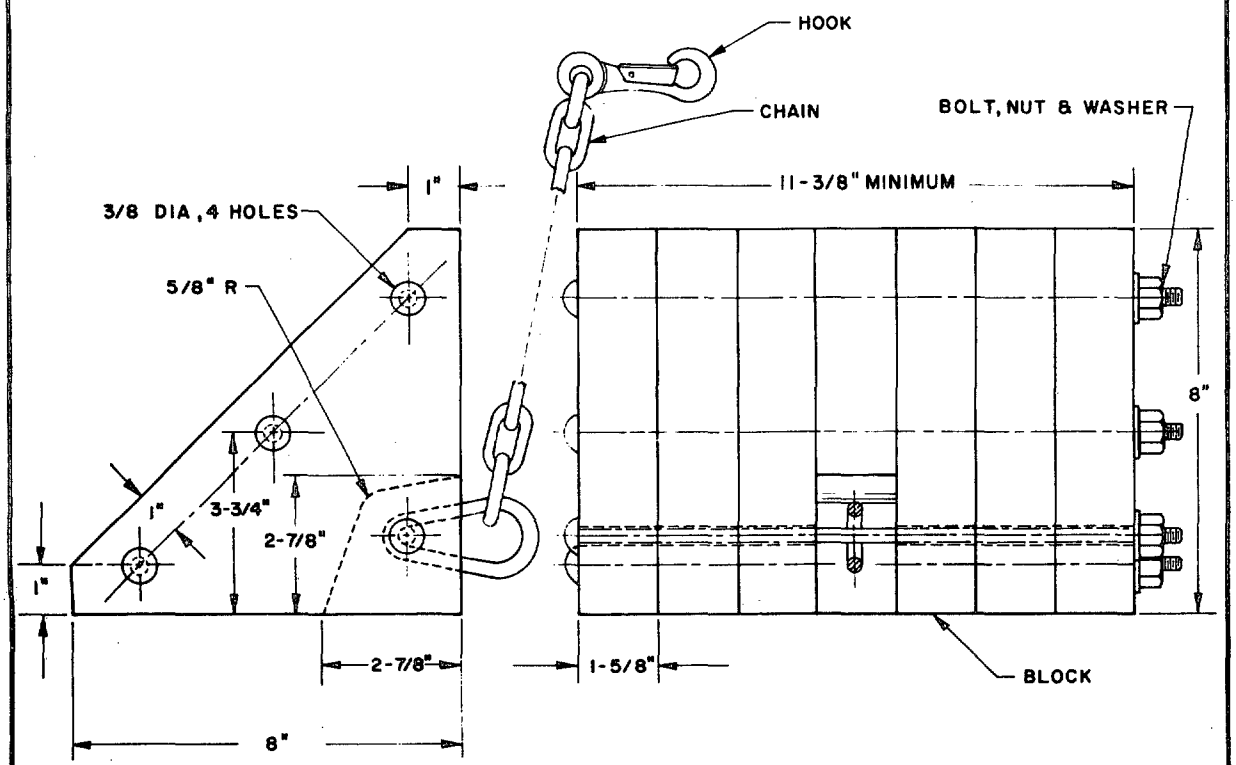
TYPICAL KEYWAY CONTOUR

FILLET RADIUS AT BOTTOM OF KEYWAY	
H	X
UNDER 1/4	SAE
1/4 , 5/16	3/32
3/8	1/8
7/16 , 1/2	5/32
9/16	3/16
5/8 , 11/16	7/32
3/4	1/4
13/16 , 7/8	9/32
15/16	5/16
1 , 1 1/16	11/32
1 1/8 TO 1 1/4	3/8

FIGURE 1. KEYS AND KEYWAYS

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CHAIN DETAILS

CHAIN WELDED, CLOSE STRAIGHT LINK, GALV'D, 1/4" NOMINAL SIZE OF CHAIN BARS.

LENGTH OF 100 LINKS = 100 INCHES

MINIMUM TENSILE STRENGTH = 3,400 LBS.

WEIGHT/100 FT. = 70 LBS.

MATERIAL NOTE

WOOD PARTS, FABRICATED, HARDWOOD.

FIGURE 2. CHOCK BLOCK ASSEMBLY

DX-603

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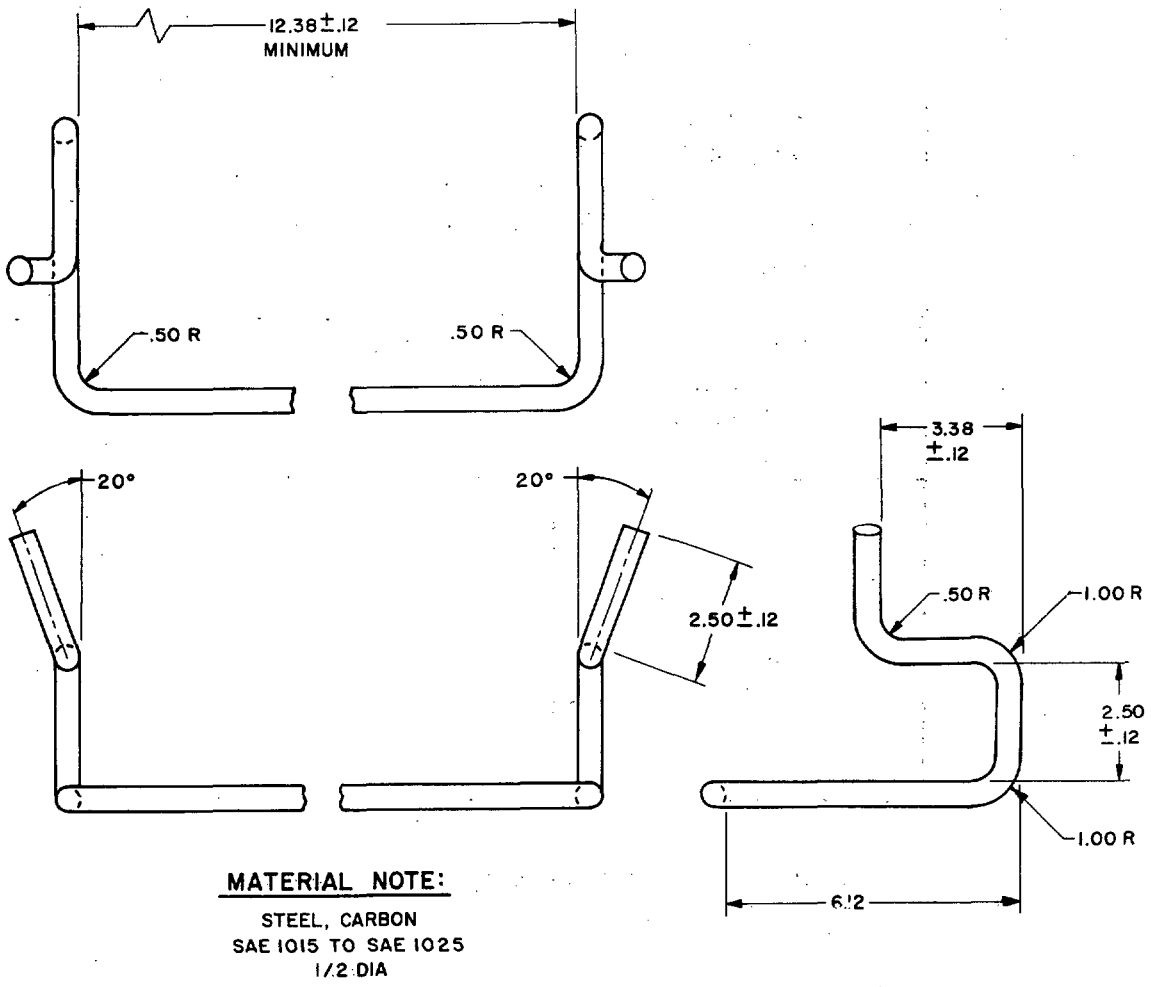


FIGURE 3. CHOCK BLOCK CRADLE

DX-604

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PRODUCTION CAPACITY

THE FOLLOWING PRODUCTION FIGURES ARE BASED ON CRUSHING QUARRY RUN ROCK WITH 80% REQUIRING CRUSHING. CAPACITIES SHOWN MAY VARY AS MUCH AS PLUS OR MINUS 25% DEPENDING ON THE TYPE OF ROCK.

MAXIMUM FEED SIZE 12" x 22"

PRODUCT SIZE	CAPACITY T.P.H.
1"	
1 1/2"	
2"	
2 1/2"	
3	

FIGURE 4. PRODUCTION CAPACITY PLATE

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